Co-ordinate Geometry

1. Reduce the equation $32x^2+52xy-7y^2-64x-52y-148=0$

2. Test the nature of the conic $x^2+12xy-4y^2-6x-4y+9=0$ and find the centre of the conic. Also reduce the equation to the standard form.

Change of axes

1. Determine the equation of the curve $2x^2+3y^2-8x+6y-7=0$ when origin is transffered to $(2,-1)$

2. Determine the equation of the curve $x^2-2xy+y^2-2x-4y+3=0$ after rotating the axes through an angle $45°$.

3. Transform the equation $11x^2+24xy+4y^2-20x-40y-5=0$ to rectangular axes through the point $(2,-1)$ and inclined at an angle $\tan^{-1}4/3$

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